

FinQuery

A local-first personal-finance analyst. Drop your bank statements into a chat, ask in your own words, get numbers you can audit.

 FinQuery

+ New chat

Dashboard

Transactions

Imports

Memory

Recent

The sample year

The sample year

Dashboard

Every figure here comes from a query this page just ran. Ask for a chart in a chat; the ones worth keeping land here.

This month

Last 3 months

This year

All - 01.01.2025 to 28.12.2025

Any date

Spent

2.346,34 €

December 2025

-73,80 € (-3 %) vs last month

-72,69 € (-3 %) vs 6-month average

Income

2.850,00 €

December 2025

0,00 € (0 %) vs last month

0,00 € (0 %) vs 6-month average

Net

503,66 €

December 2025

+73,80 € (+17,2 %) vs last month

+72,69 € (+16,9 %) vs 6-month average

Needs review

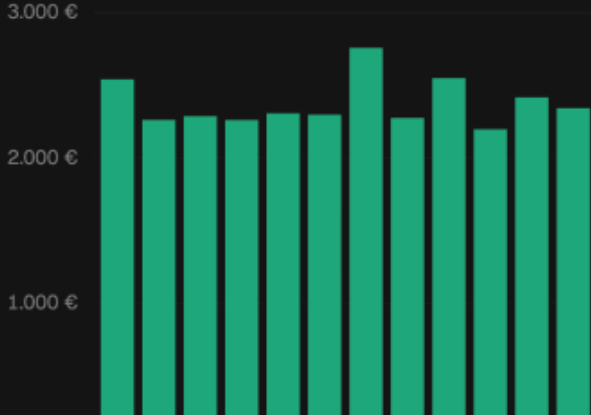
25

01.01.2025 to 28.12.2025

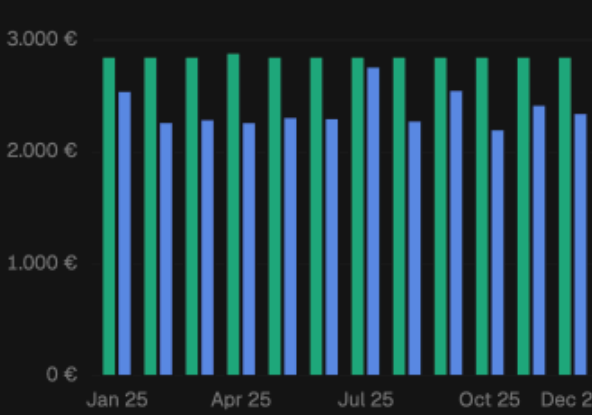
Bookings without a category

Answer them in a chat

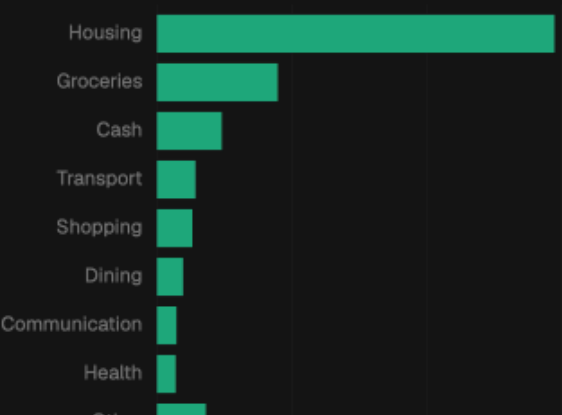
Spending per month



Income against spending per month



Top categories



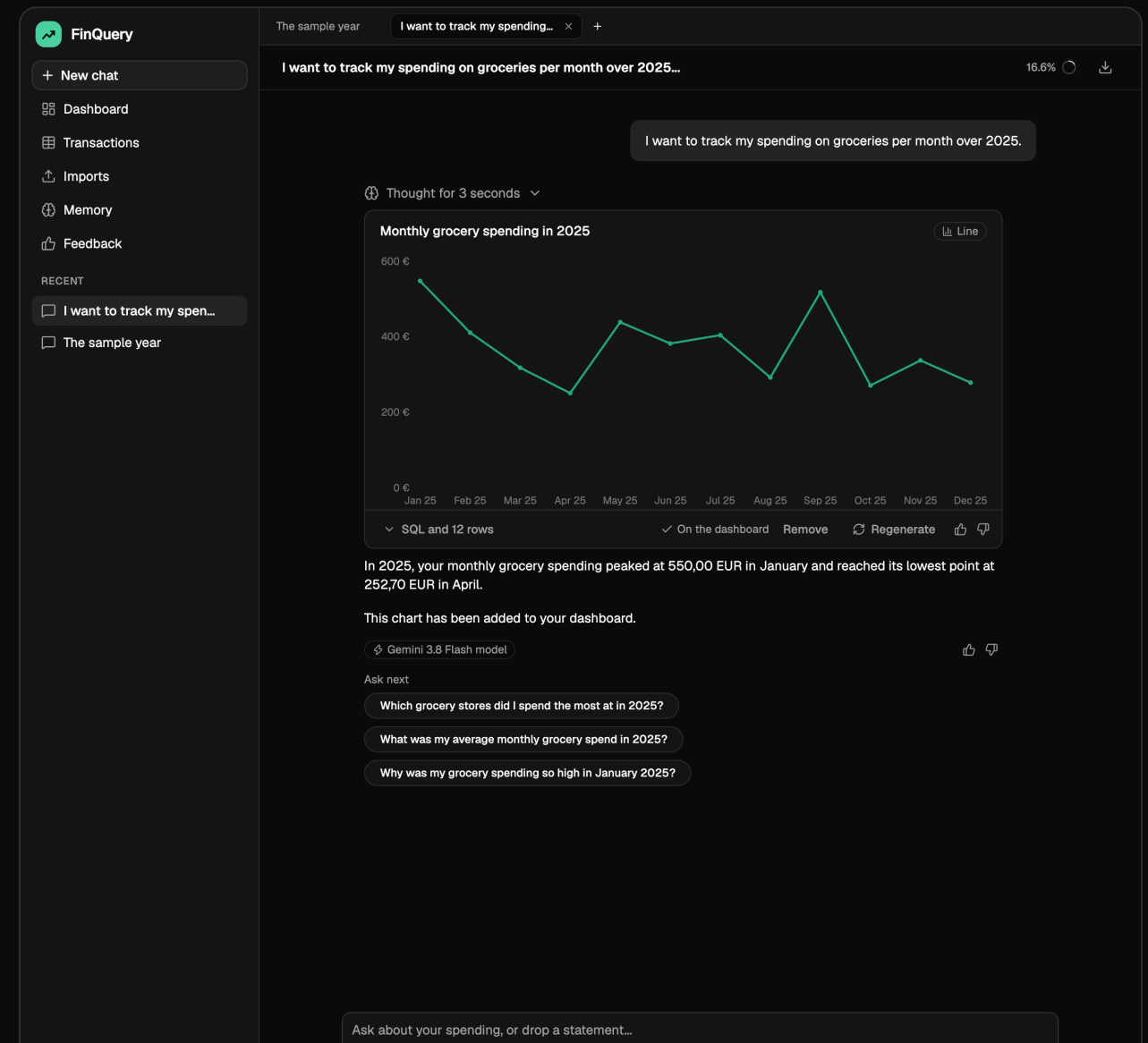
Rahul Singh and Liudvikas Zekas, Hasso Plattner Institute, Potsdam

Everything runs on the laptop: two Gemma 4 models through llama.cpp, SQLite, a React app

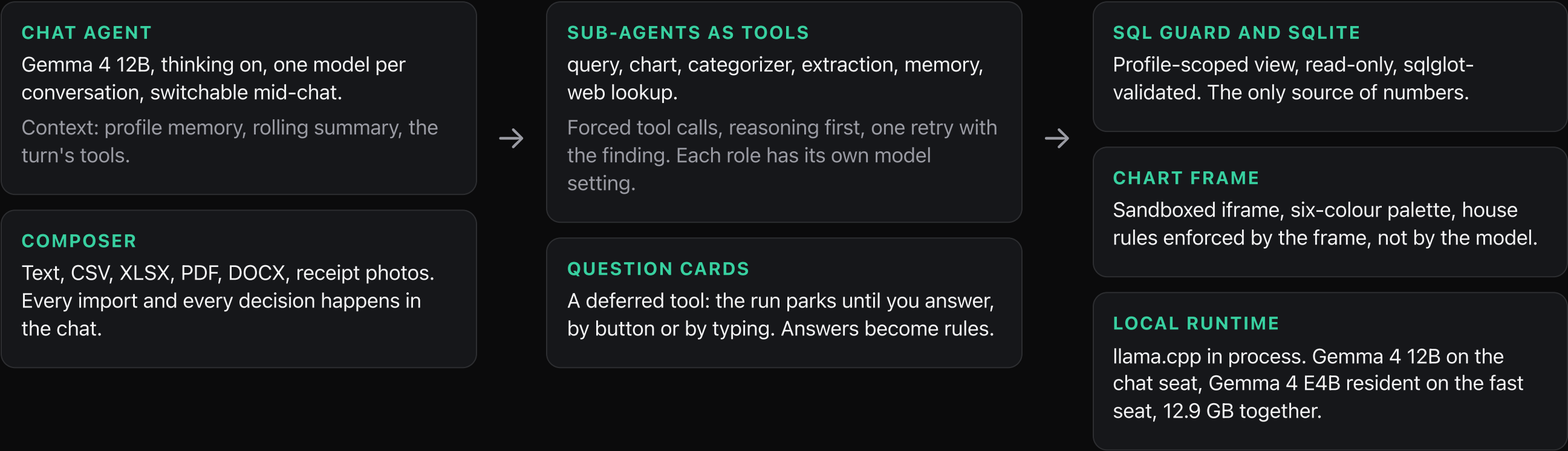
Numbers come from executed queries, never from the model

Every figure in an answer is the result of a **SQL statement the assistant really ran** against your own transactions, shown next to the answer. The model writes the question and the words. The database writes the numbers.

- **A guard in front of the database.** One SELECT over a profile-scoped view, parsed with sqlglot, read-only, capped, refused when it invents categories or leaks cents.
- **A check pass after the query.** A degenerate result or a wrong-shaped answer is sent back once with the finding. 69 to 79 percent figure match on the same model.
- **Prose is verified too.** A figure typed into the answer that no query produced is flagged.
- **Charts are checked code.** The chart sub-agent writes JavaScript for TanStack Charts; a self-check runs it on the real rows before a sandboxed frame draws it.



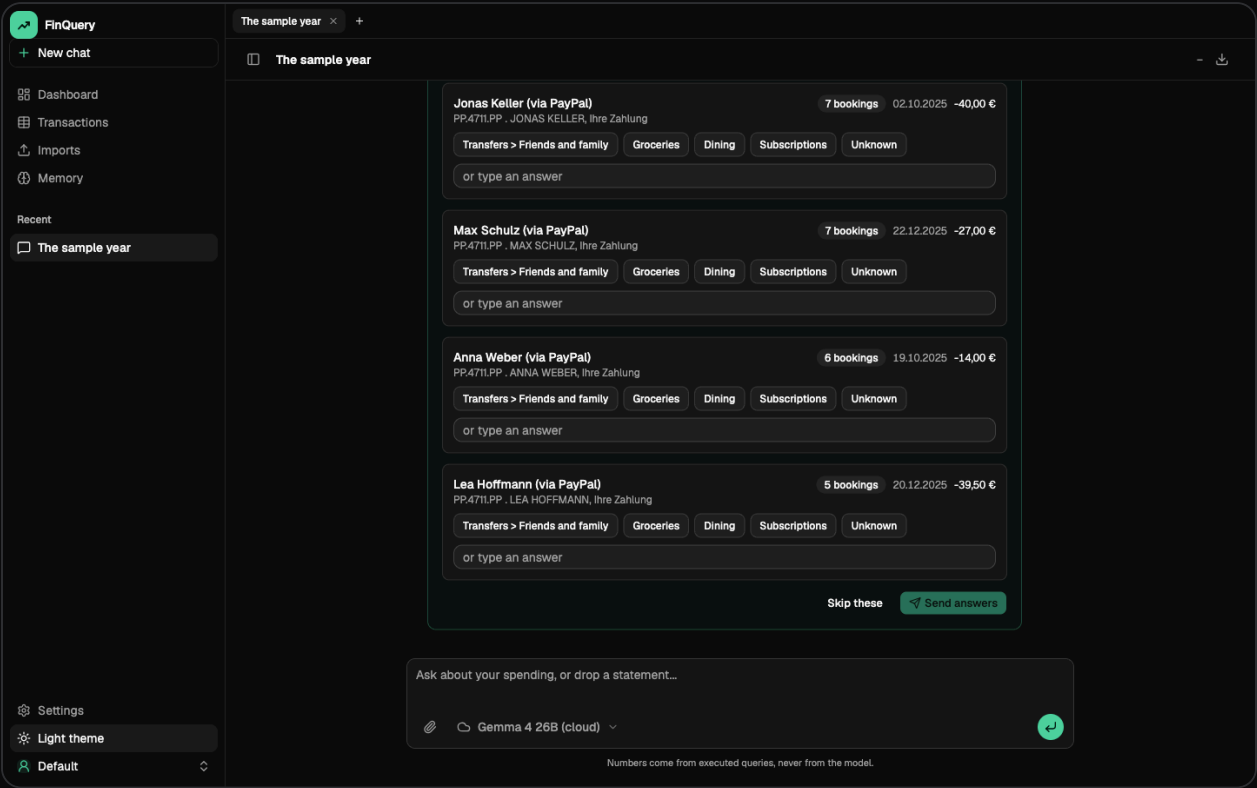
One chat agent, tools, sub-agents, a guard, a sandbox



Intelligent context management

compression, isolation, selection

- **Isolation.** A profile is a workspace: its own transactions, conversations, memory, categories and rules. Every query runs through a temp view scoped to the profile.
- **Selection.** Each turn assembles exactly what it needs: the distilled memory block, the rolling summary, the recent turns, and only the tools that apply (web lookup is not even declared when it is off).
- **Compression.** At 60 percent of the 32k budget the older turns fold into a summary written by the fast model; the divider in the transcript is editable, so the user owns what was kept.
- **Memory.** After every turn at most two durable facts are distilled, never figures or dates unless the user states a rule. A Memory page shows and edits them.



The header badge shows tokens used against the budget; the divider marks where the summary starts.

Sub-agent deployment

an LLM calls another LLM to do one task

- **Six sub-agents, each a tool call.** The chat agent decides when; the sub-agent gets one prompt, a forced tool schema and a reasoning field before the answer.
- **Self-correcting.** The query sub-agent gets a check pass and one rewrite; the chart sub-agent gets a self-check on the real rows and visible repair rounds.
- **Narrated.** Every sub-agent step streams into the transcript as a chain of thought, so the user sees what was asked and answered.
- **One model setting per role.** Default: the conversation's chat model. Switch a role to the fast seat and it runs on E4B, where the adapters attach.

Not claimed: the extra credit for a controller that continues without waiting. The turn is a server task that survives a closed tab, but the chat agent waits for its tool.

FinQuery

+ New chat

Dashboard

Transactions

Imports

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Recent

Your conversations will appear here.

Settings

Default

chat model

Qwen3.5 9B (local)

local;qwen3.5-9b

on disk

chat model - 32k context - loads on first use

✓ Qwen3.5-9B-Q4_K_M.gguf

weights

5.68 GB

✓ mmproj-F16.gguf

projector

0.92 GB

Qwen3.5 9B (cloud)

openrouter;qwen/qwen3.5-9b

cloud

chat model

Gemma 4 E4B

local;fast

on disk

sub-agent fast slot - 32k context - loads on first use

✓ gemma-4-E4B-it-Q4_K_M.gguf

weights

4.98 GB

✓ mmproj-F16.gguf

projector

0.99 GB

Gemma 4 26B

openrouter;fast

cloud

sub-agent fast slot

Sub-agents

Which model each job behind a tool runs on, for a chat on openrouter:google/gemma-4-26b-a4b-it. A chat on another entry takes every role set to chat or fast with it.

query	chat	Gemma 4 26B (cloud)
chart	chat	Gemma 4 26B (cloud)
categorizer	chat	Gemma 4 26B (cloud)
extraction	chat	Gemma 4 26B (cloud)
memory	chat	Gemma 4 26B (cloud)
summary	chat	Gemma 4 26B (cloud)
weblookup	chat	Gemma 4 26B (cloud)

Adapters

query

no file at /Users/chichurita/Dev/finquery/models/adapters/query.gguf, a role on the fast slot runs on the base weights

chart

no file at /Users/chichurita/Dev/finquery/models/adapters/chart.gguf, a role on the fast slot runs on the base weights

Web lookup

Lets the assistant search the web for a merchant it does not recognize. Only a scrubbed merchant name is ever sent: no amounts, no dates, no account numbers, and a booking whose merchant reads as a person is refused. Every request is logged below before it is sent.

Off

Outbound log

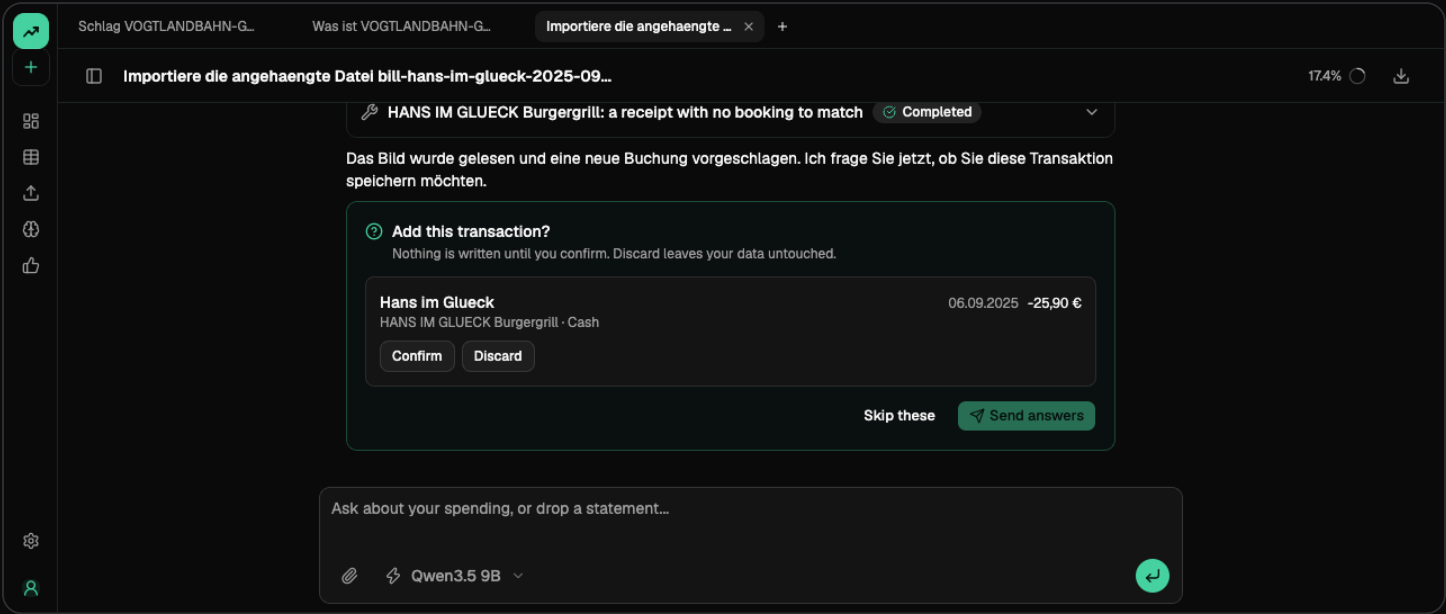
nothing has ever left this machine for this profile

The Settings models card: the four catalog entries, both fast seats and the seven roles.

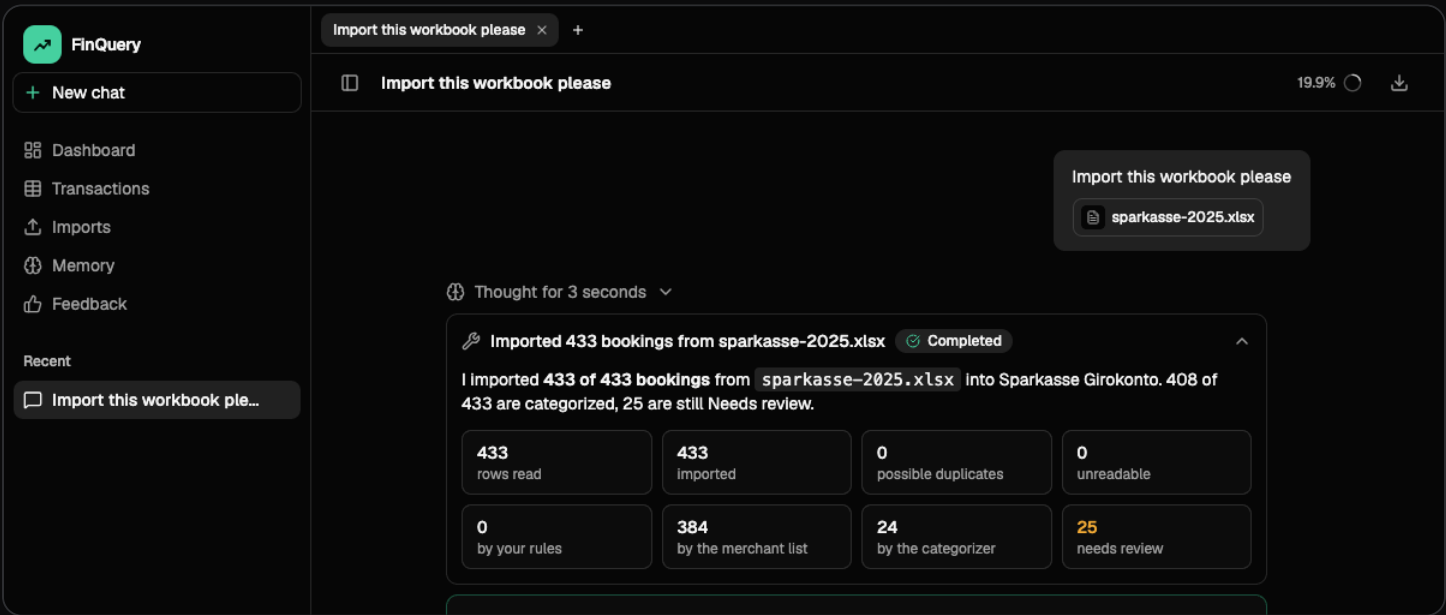
Multimodal ingestion

PDF, images, text, CSV, XLSX, DOCX

- **Inherently multimodal models.** Gemma 4 E4B and 12B read scanned statements and receipt photos through their vision projectors. Text-layer PDFs are read as text first, vision only where there is none.
- **Two guards on every extracted figure.** Verbatim: every amount and date must occur in the source.
Reconciliation: opening balance plus bookings equals closing balance. Anything flagged goes to a review card, nothing is written silently.
- **Measured.** 19 of 19 receipt totals right on real and web receipts; the synthetic statement PDF reconciles to the cent; the real Edeka receipt needed an EXIF rotation fix.
- **Excel and Word** go through the same paths as CSV and pasted text. Audio is out of scope and said so.



A receipt photo becomes a draft with title, category and item legs.

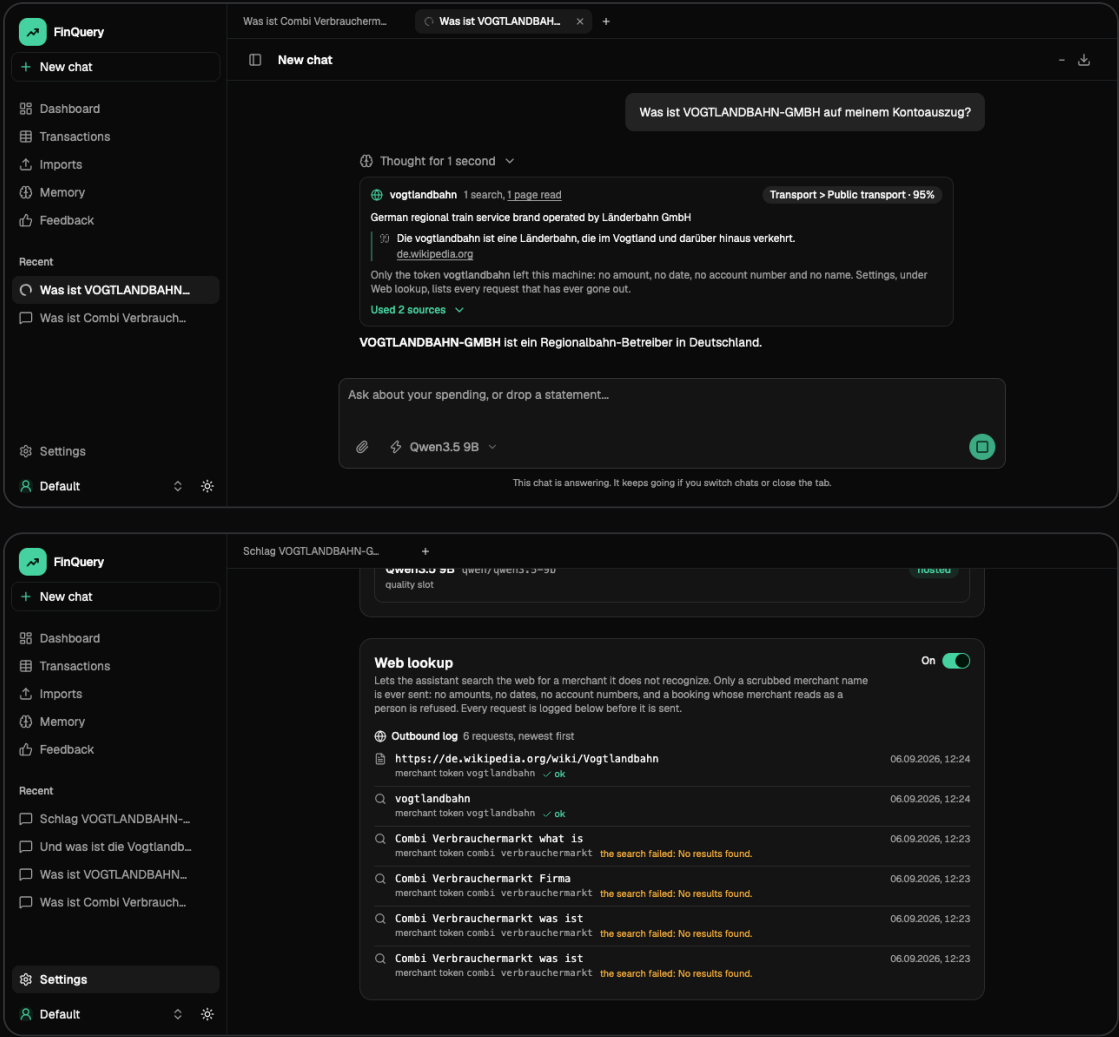


An Excel export imported through the chat: 433 rows, the preset recognized without a model call.

Self-controlled web search

find a page, read it, quote it

- **A loop the model drives.** Search, fetch or finish, up to four searches and three page reads, with a floor in code: at least one search, and a page read whenever a hit is the merchant's own site or Wikipedia.
- **Evidence, not vibes.** The finish carries a verbatim quote that must occur in what was read; confidence is capped when no source names the merchant.
- **Privacy by construction.** Only a scrubbed merchant token leaves the machine: never an amount, a date, an IBAN or a person's name. Every request is journaled before it is sent and shown in Settings.
- **Shared by three callers.** The chat tool, the categorizer at import in a batch, and receipt headers. Measured yesterday: lookups without a search 9 to 0, pages read 0 to 12.



The card and the outbound log: token, search, page, quote.

Which local models, and the fine-tuning loop

MODEL, Q4_K_M	SQL FIGURE MATCH	CHART FIGURE MATCH	END TO END	LAPTOP TOK/S
Gemma 4 E4B	66 %	45 %	53 %	47
Qwen3.5 9B	70 %	42 %	73 %	30
Gemma 4 12B	87 %	81 %	77 %	22

152 SQL and 73 chart cases, held-out third never used for training. Same GGUFs and wire formats as the laptop, one RTX PRO 6000 per job. [bench/results/20260906-cluster-compare.md](#)

- The decision rule was written before the numbers: accuracy first, speed within five points, the pair must fit in 13.5 GB. Result: Gemma 4 12B for chat and sub-agents, E4B resident as the fast seat.
- Four QLoRA adapters in progress query and chart on both bases. Training data written by 60 agents over five new synthetic households, kept only by execution: guard, database, an independent second statement, chart self-check, a rendered picture judged.
- The benchmark is the validation set. Frozen split, the shipped household is benchmark-only and unseen in training, checkpoints every epoch scored on a quick set, only the best is converted to GGUF and scored for real.
- Smoke-tested end to end on an H100 yesterday: 20 steps, conversion, adapter attached, five benchmark cases right.

WHAT WE OWN

Decisions we made, and the ones we reversed on evidence

KEPT

Numbers only from executed SQL. Sub-agents as forced tool calls with reasoning first. Every import and every decision in the chat. Training equals production: a sample is byte for byte the prompt the sub-agent sends.

NOT CLAIMED, ON PURPOSE

Sub-agent extra credit. Python code execution (our sandbox runs JavaScript charts). Audio ingestion.

Docs: docs/explainers (one chapter per required bullet and per elective, with sources), docs/adr (13 decisions), bench/ (the benchmark and every run), training/ (data harness and cluster loop).

REVERSED

Qwen3.5 9B as quality slot, chosen on memory, replaced by Gemma 4 12B after the cluster benchmark. Preference optimisation built, measured and removed for multimodal ingestion. "No dashboard" turned into a dashboard of guarded queries with a date range.

NEXT

Adapter numbers against vanilla, the video, a second data round if the learning curve asks for it.